Mixed applications

1

a $h = 86.52$ **b** $a = 20.37$ **c** $a = 10.00$ **d** $y = 2.4$

2 $l = 20.6 \text{ m}$

3

a $x = 149.12^\circ$ **b** $\theta = 102^\circ$ **c** $\theta = 146^\circ$

4

a $\theta = 37.22^\circ$ **b** $d = 3.4 \text{ cm}$

5 $x = 22^\circ$

6

a $AE = 13.8577$ **b** $BE = 7.6433$ **c** $AB = 21.50$ **d** $BD = 6.3$

7

a $73\,500 \text{ m}$ **b** $h = 41\,101 \text{ m}$

8 $h = 4.4 \text{ m}$

9

a 74 **b** $d = 28.35$ **c** $h = 16.66$

10

a $\angle ACB = 102^\circ$ **b** $b = 226.55 \text{ m}$ **c** $h = 117 \text{ m}$

11

a $x = 23.88 \text{ m}$ **b** $k = 25.95 \text{ m}$ **c** 65.95 m

12

a 21.89° **b** 123.11° **c** $AB = 29.2$

13

a 136° **b** $SW = 234 \text{ km}$ **c** $h = 102.58 \text{ km}$

14

a 23° **b** $a = 35.65$ **c** $h = 20.4$

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15
a 3799 km **b** 4915.5 km **c** $x = 4014.7\text{ km}$

16 $\theta = 33.56^\circ$

17 $x = 37.79\text{ km}$

18
a $x = 48^\circ$ **b** 132°

19
a $\angle ABC = 90^\circ$ **b** $x = 27^\circ$ **c** 308°T

20
a $\frac{3}{2}x^2\text{ cm}^2$ **b** $\frac{1}{4}$

21
a $\theta = 106.3$ **b** $x = 36.9$ **c** $d = 12.0$

22
a $\angle ASC = 90^\circ$ **b** $AC = 2222\text{ km}$ **c** $x = 36^\circ$ **d** 195°T

23
a $x = 26.73\text{ km}$ **b** $b = 20^\circ$ **c** $\text{N } 34^\circ\text{W}$

24 $x = 41$

25 $y = 2\sqrt{7}\text{ cm}$

26
a 47.13 m **b** 76.3 m **c** $\theta = 23^\circ$

27
a $\theta = 52.6^\circ$ **b** 136.66 cm^2

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